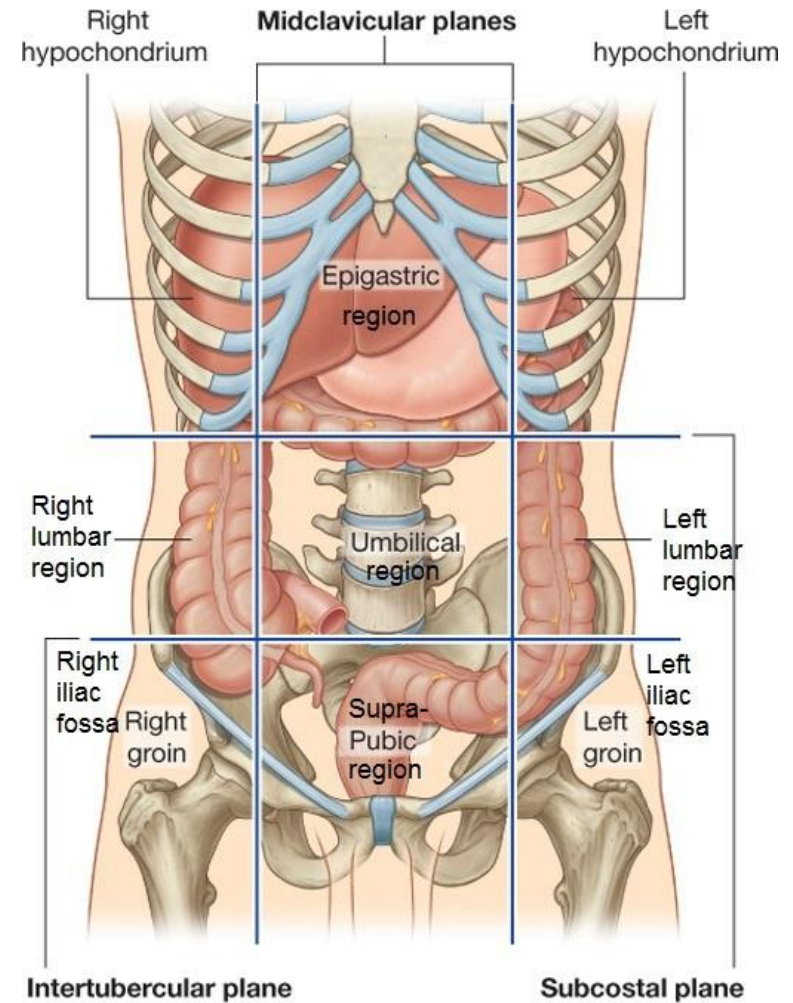


# WHAT IS DISCUSSED IN THE LECTURE?

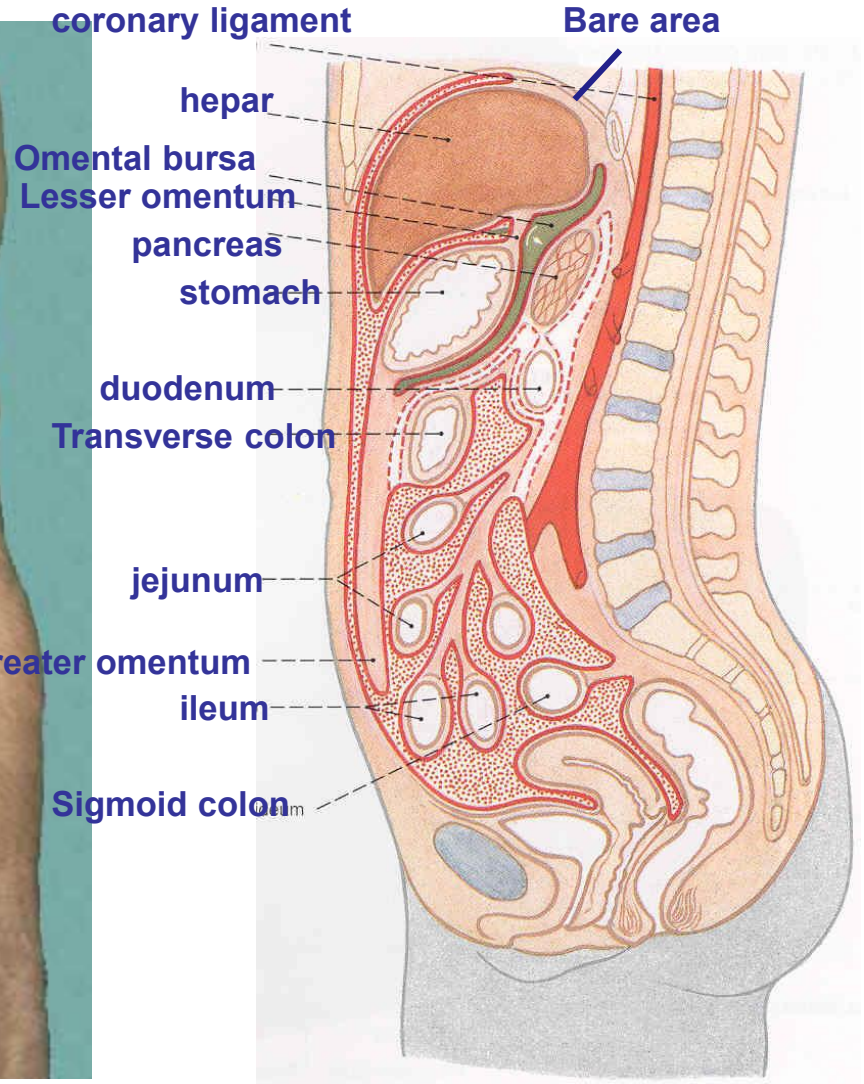
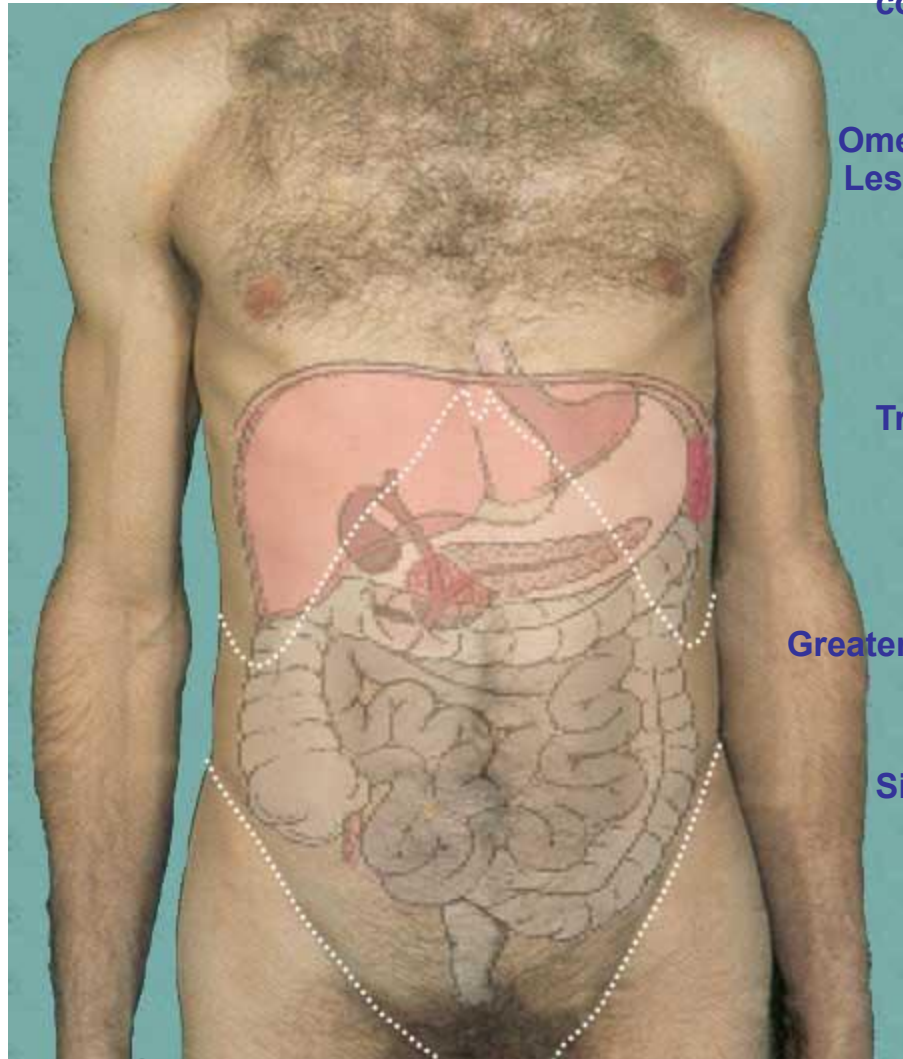
- **Position of the large intestine**
- **How can we distinguish the large and small intestines?**
- **Divisions of the large intestine + Peritoneal relations**
- **Blood supply + Anastomoses**
- **Innervation**
- **Lymphatic drainage**



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# ***POSITION OF THE LARGE INTESTINE AND RECTUM***



# ANATOMICAL DIVISIONS

**CECUM + vermiform appendix**

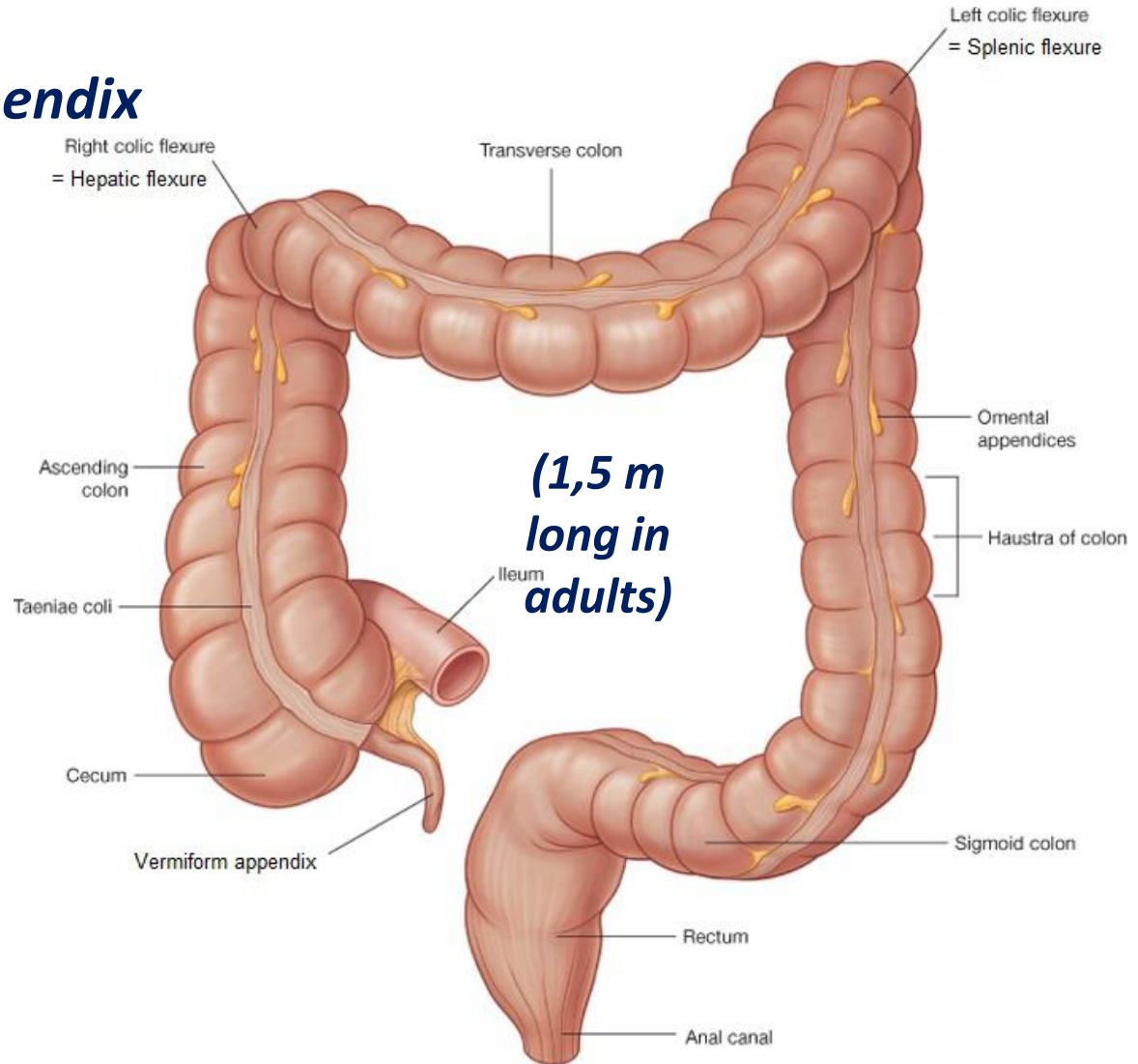
**ASCENDING COLON**

**TRANSVERSE COLON**

**DESCENDING COLON**

**SIGMOID COLON**

**RECTUM**



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# HOW CAN WE DISTINGUISH THE LARGE AND SMALL INTESTINES?

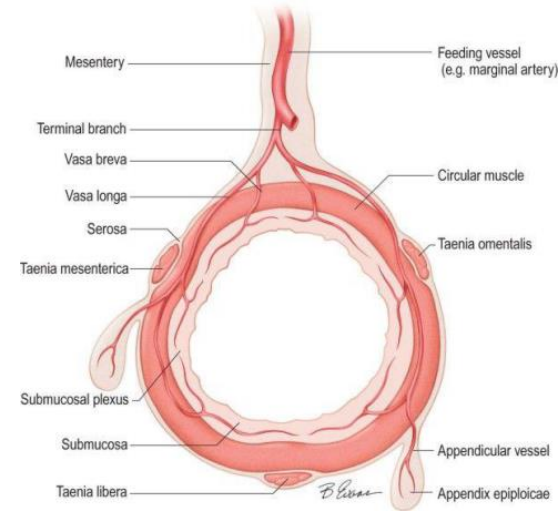
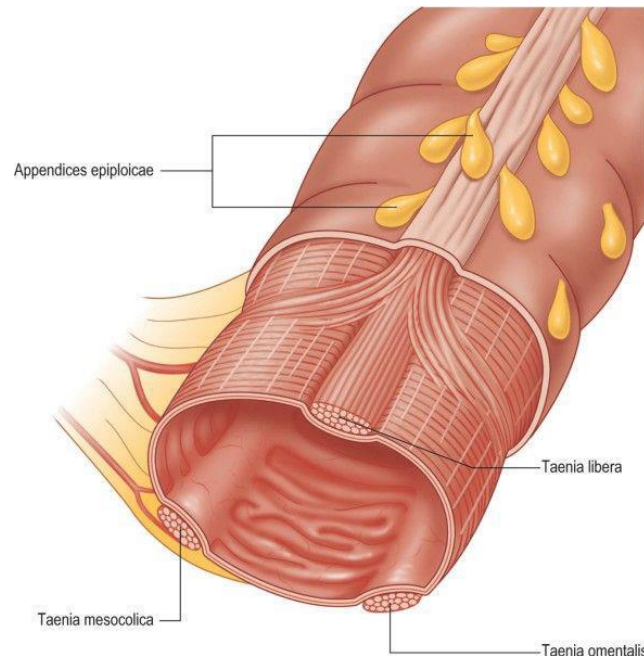
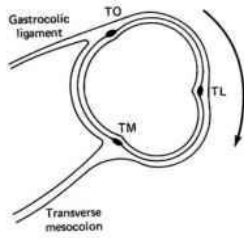
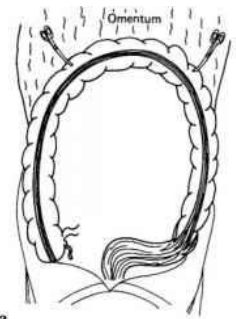
## External features

**Epiploic appendices** - protrusions of the peritoneum filled with areolar /fatty connective tissue

**Haustra** (plural) - saccular dilations of the wall

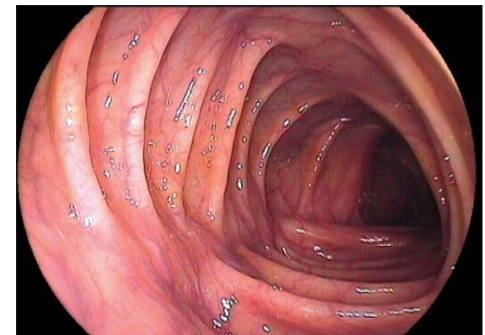
**Taeniae** (plural) - strong bands of the tunica muscularis (absent from the vermiform appendix, sigmoid colon and rectum)

**Omental**  
**Free**  
**Mesocolic**



## Internal features

**Semicircular folds**  
**No villi**



# VERMIFORM APPENDIX AND CECUM

Located in the right iliac fossa

Both parts are mostly

**INTRAPERITONAL**

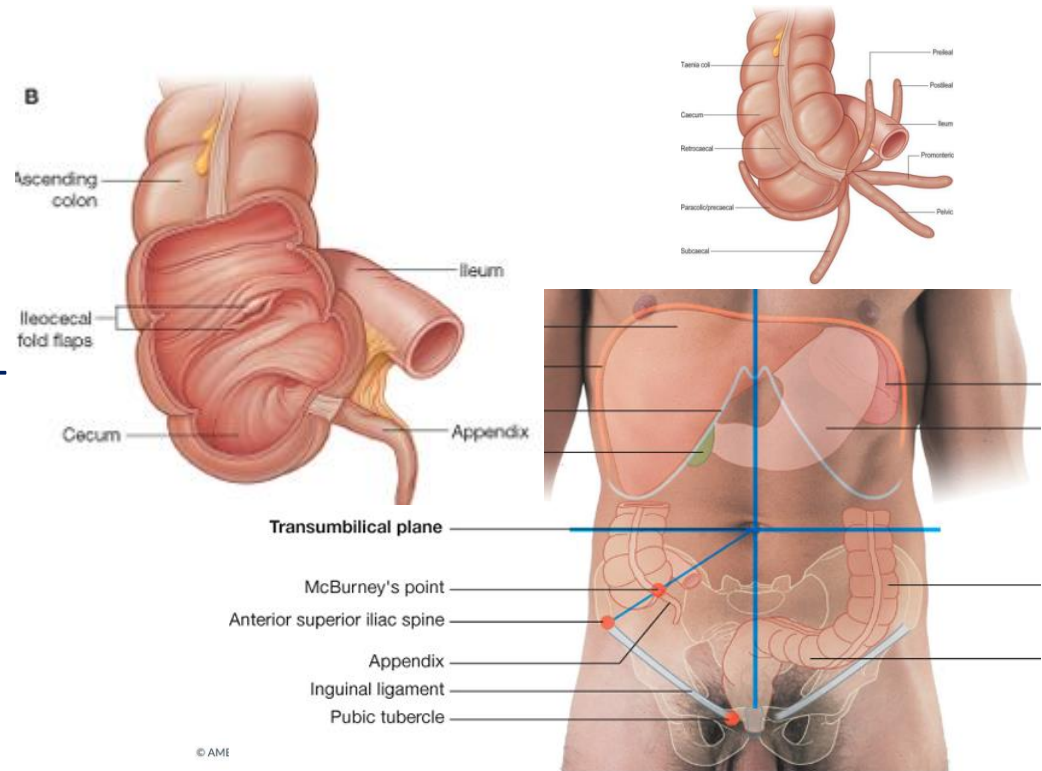
**Ileocecal recesses (peritoneum)**

**Ileocecal orifice + Valve of Bauhin** –  
two lips (longitudinal muscle layer of  
ileum)

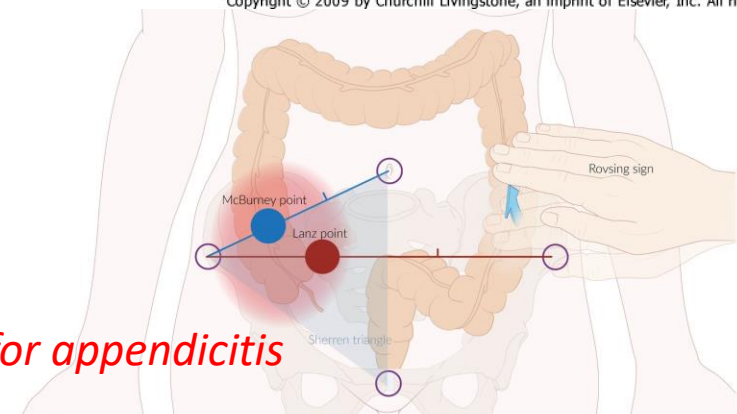
**Vermiform appendix** starts at the  
merging of the 3 taeniae (McBurney  
point) and ends at the Lanz's point  
(approx 3-20 cm long)

**RETROCECAL / RETROCOLIC**  
**Great varieties in the position**

**Mesoappendix** (if intraperitoneal)



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**Pressure points for appendicitis**

# ASCENDING COLON

Continuation of the cecum

**3 taeniae**

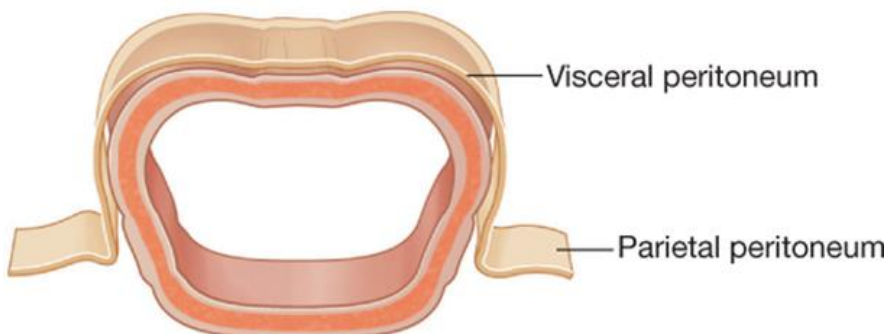
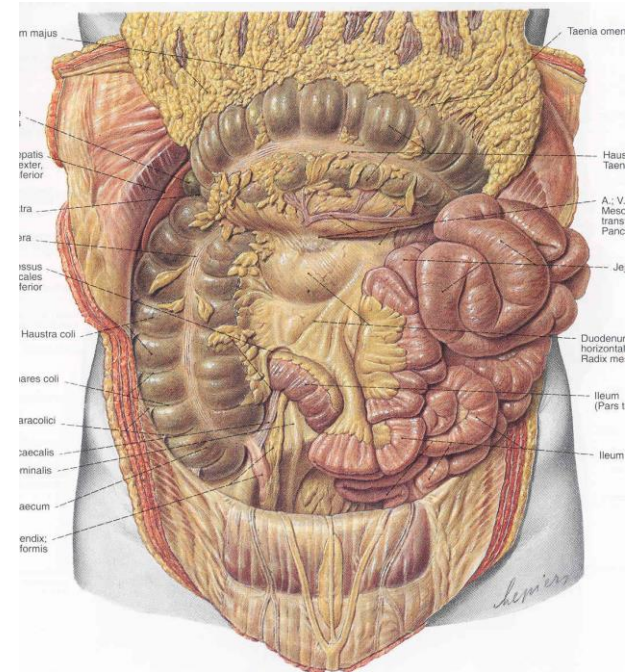
Partially intraperitoneal/retroperitoneal

**Paracolic sulcus**

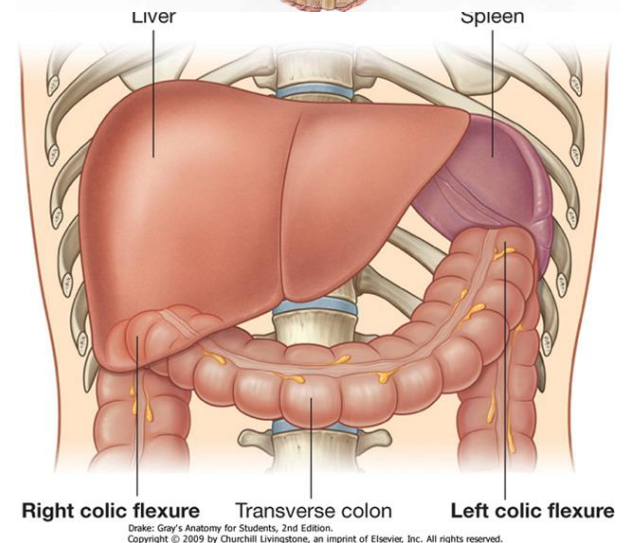
**Position:** lateral to the *midinguinal line*

- up to the *transpyloric plane*
- ends in the *right colic flexure*

**Touches** the posterior abdominal wall + right kidney



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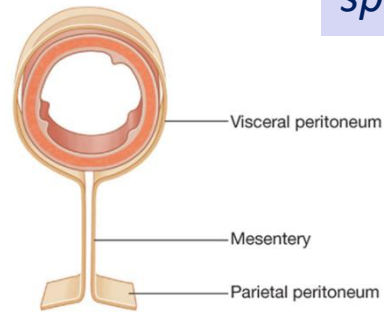


# TRANSVERSE COLON

Stretches from the **right colic flexure** to the **left colic flexure**  
(right lumbar region, across the epigastrium to the left hypochondrium)  
then continues downwards below the spleen

## 3 taeniae

Fully intraperitoneal



## *transverse mesocolon*

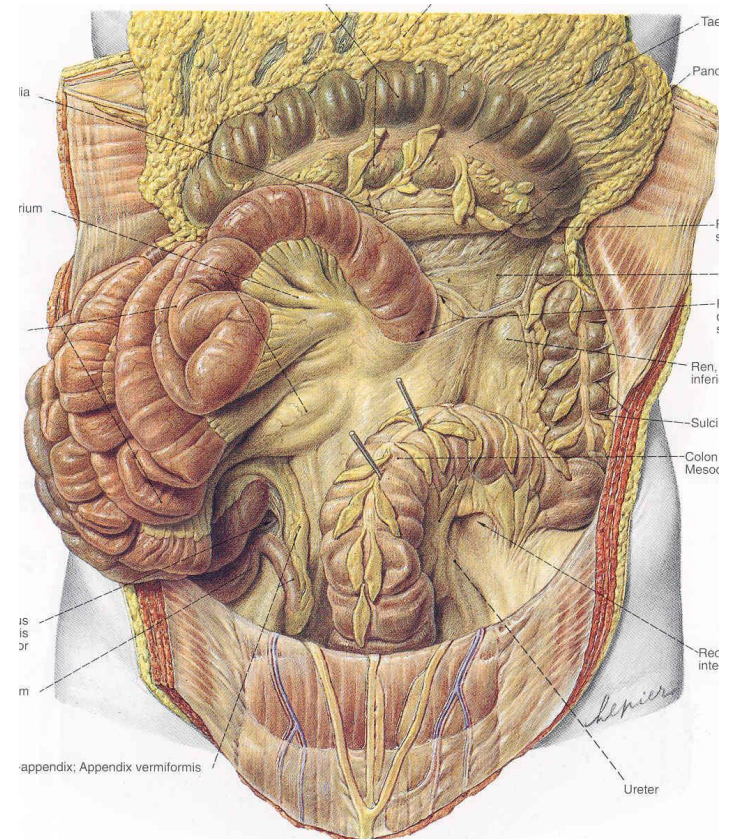
**gastrocolic ligament** (of the greater omentum)  
connects to the greater curvature  
of the stomach

## **Syntopy of the right colic flexure**

Right kidney  
right lobe of liver (colic impression)  
biliary vesicle  
duodenum (desc part)

## **Syntopy of the left colic flexure**

Lies higher than the right flexure  
phrenicocolic ligament \* 10-11th rib level  
tail of pancreas  
spleen (colic impression)



This anatomical illustration depicts the posterior abdominal wall and retroperitoneum. The pancreas is shown in the upper left, with its head and body extending towards the right. The kidneys are visible on either side of the vertebral column, with the right kidney partially obscured by the pancreas. The illustration shows the major blood vessels (aorta and inferior vena cava) and the ureters. The retroperitoneum is shown in a dissected view, revealing the underlying structures. The illustration is signed 'Leprie' in the lower right corner.

Labels on the left side of the illustration include:

- lia
- rium
- is
- is
- or
- m

Labels on the right side of the illustration include:

- Tan
- Pan
- Ren, infer
- Sulc
- Colon
- Meso
- Ren, inter
- Ureter

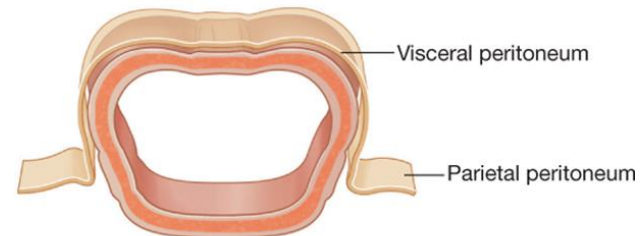
A label at the bottom left points to the appendix: "appendix; Appendix vermiformis".

Thinnest part of the large intestine  
*Develops from the hindgut*

## Partially intraperitoneal/retroperitoneal

**Position:** lateral to the *midinguinal line*  
- ends in the *sigmoid colon*

**Touches** the spleen, the posterior abdominal wall/diaphragm + left kidney





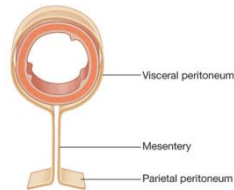
## ***SIGMOID COLON***

Descends from the left iliac fossa to the pelvis then ascends and again descends (***sigmoid looping***) to continue in the rectum at S3 level

*Develops from the hindgut*

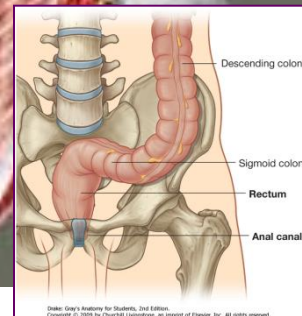
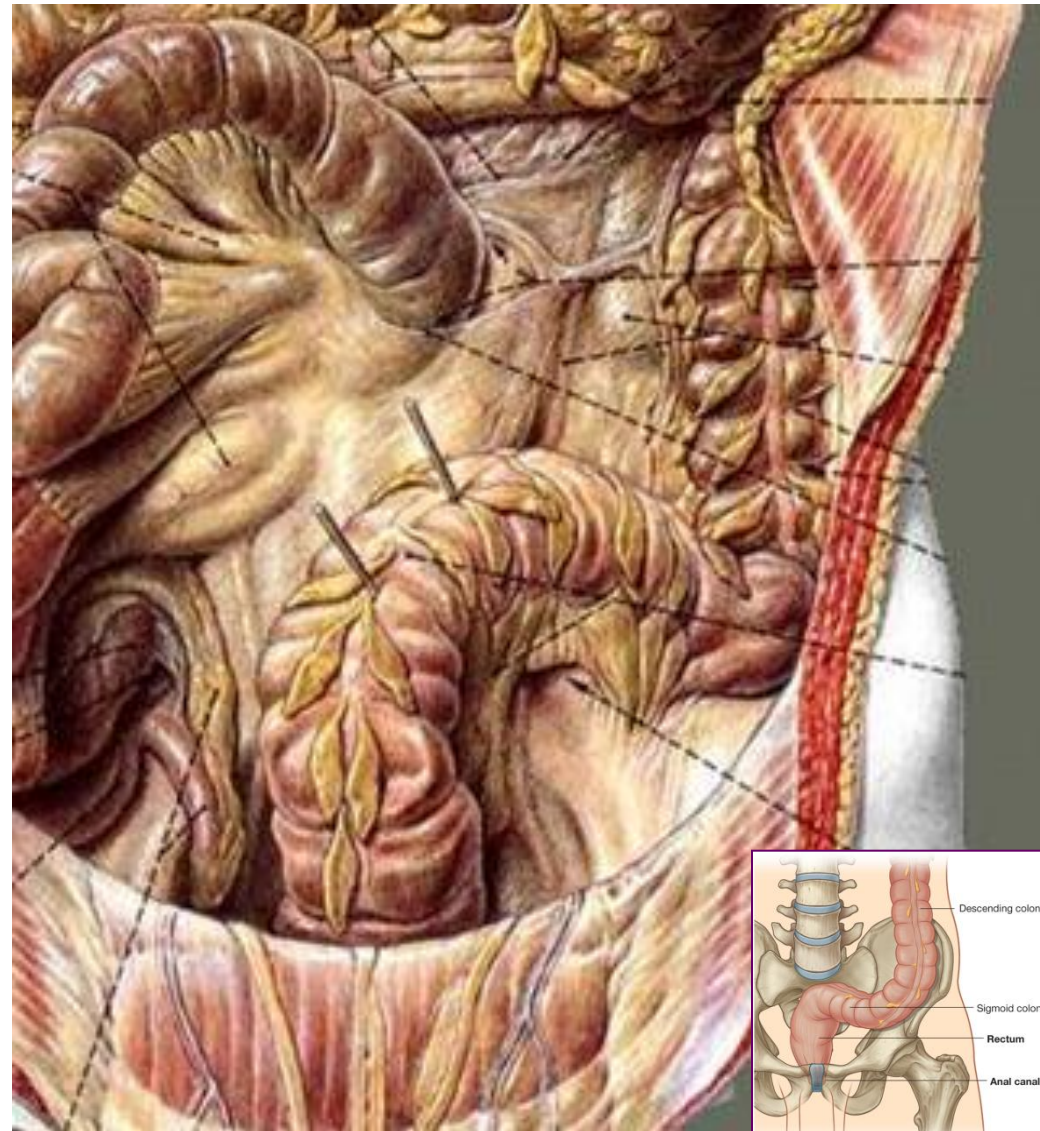
(3)  2 taeniae

**Intraperitoneal -  
mesosigmoideum  
mobile part of colon**



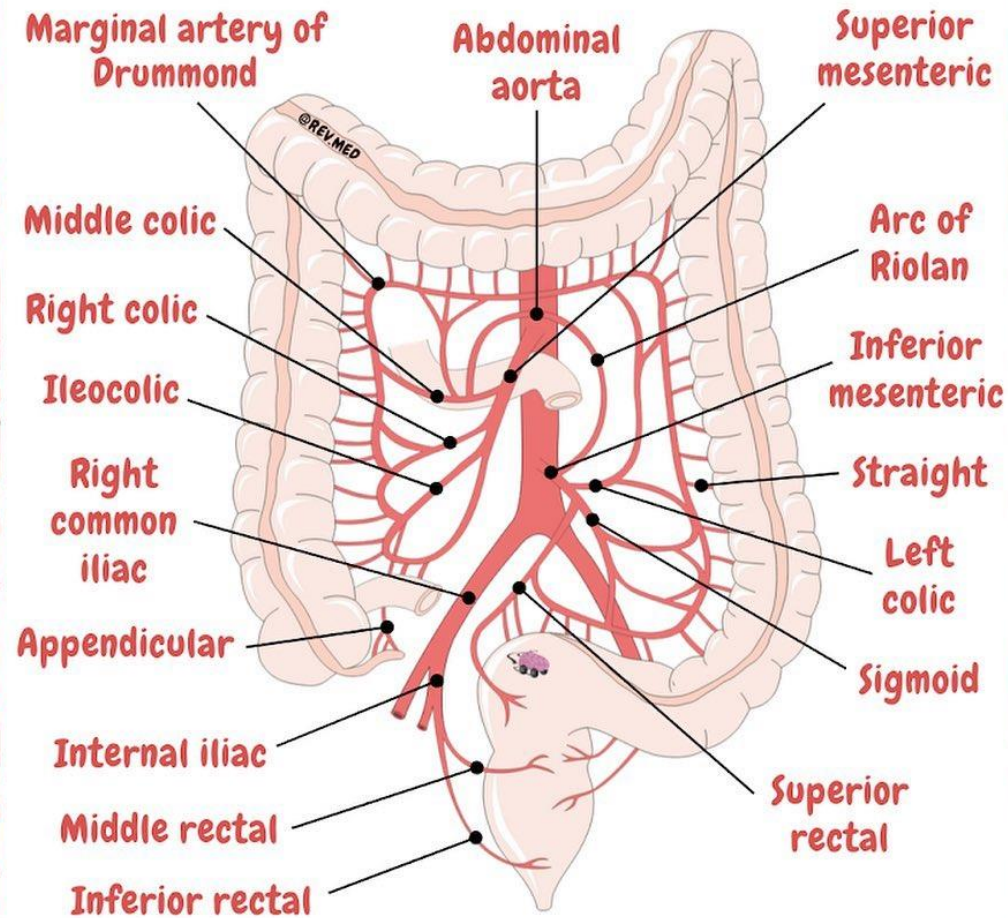
## Intersigmoid recess

**Touches** the loops of the small intestine and the pelvic organs (M: bladder /rectum  
F: uterus/rectum)



# Large Intestine

## BLOOD SUPPLY





# ***SUPERIOR MESENTERIC ARTERY (L1)***

## ***Ileocolic artery***

Supplies the cecum, vermiform appendix and the beginning of the ascending colon.

- *appendicular artery*
- *cecal arteries*

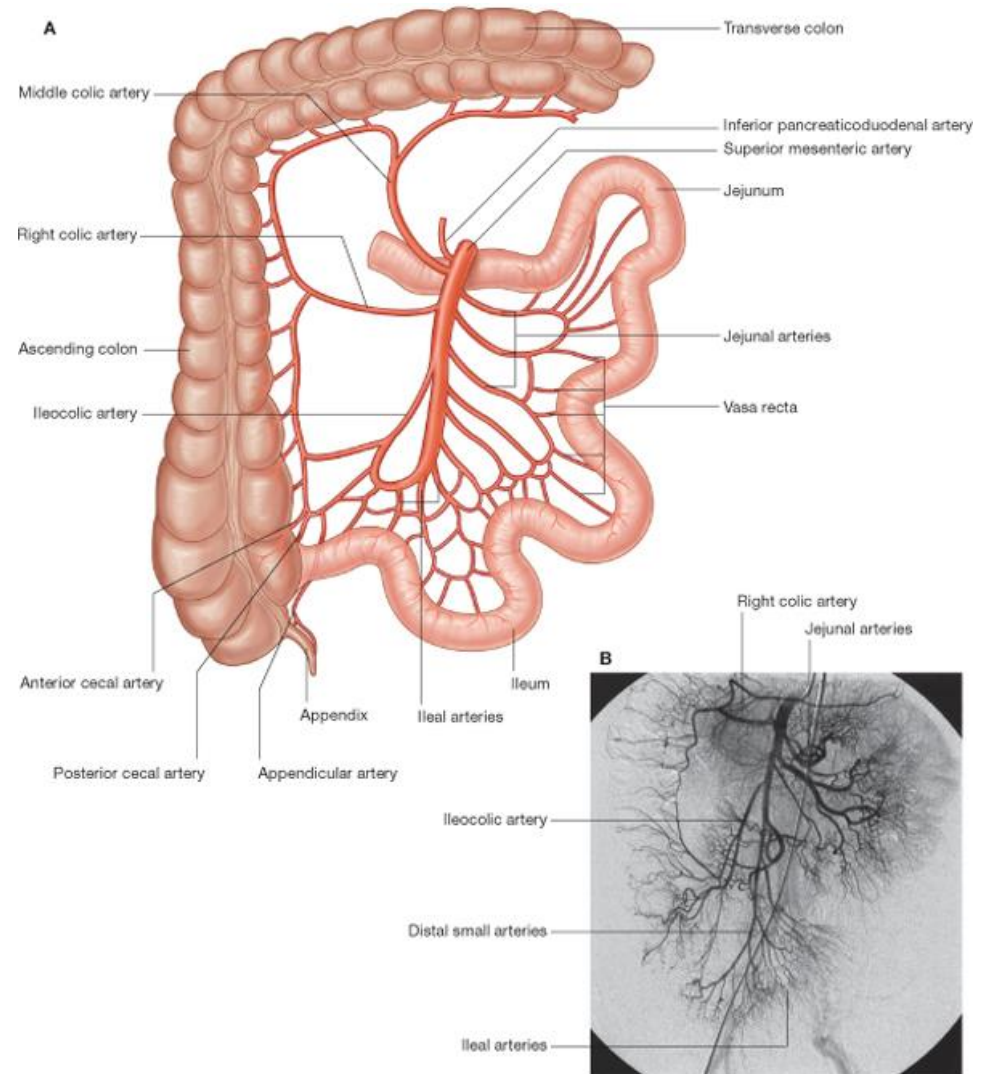
## ***Right colic artery***

Supplies the ascending colon and the right colic flexure

## ***Middle colic artery***

Supplies the right colic flexure and the transverse colon

***(developmental border between midgut and hindgut)***



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# INFERIOR MESENTERIC ARTERY (L3)

## **Left colic artery**

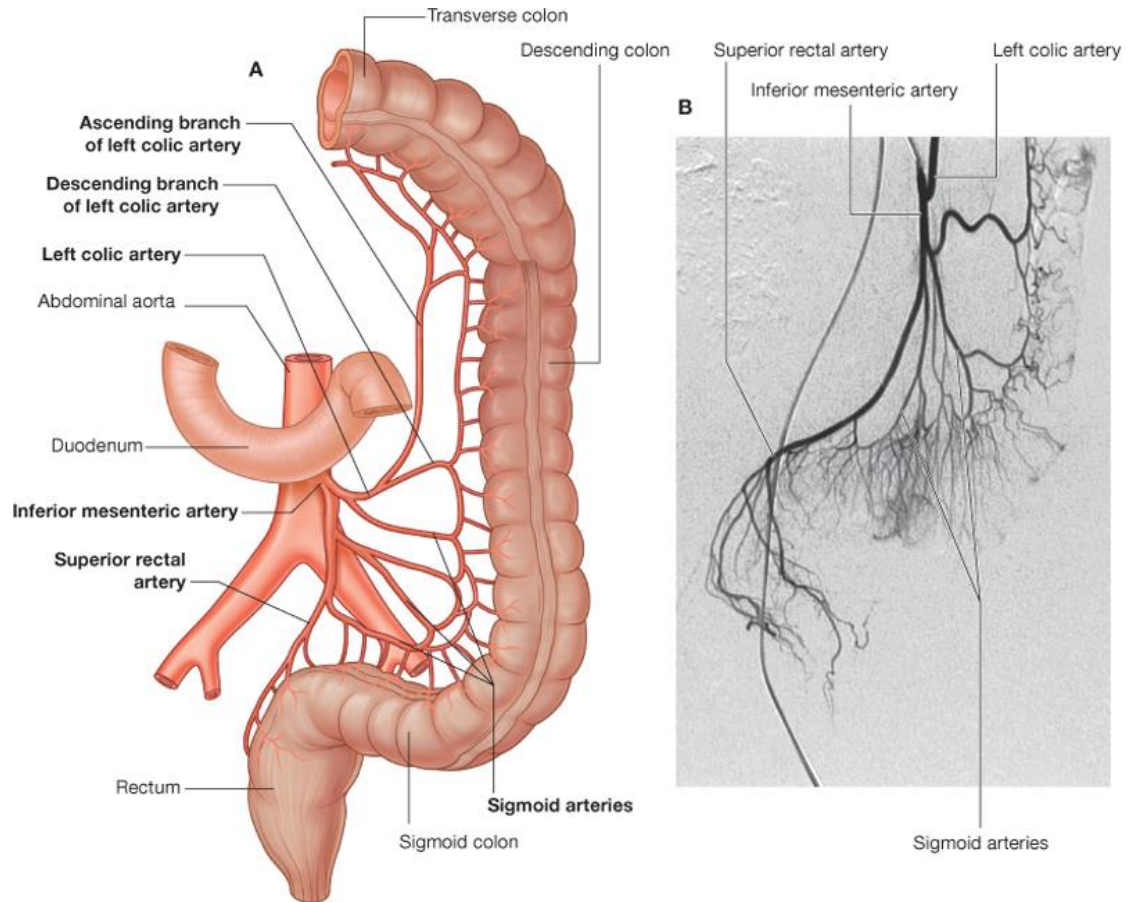
supplies part of the transverse colon, the left colic flexure and the descending colon

## **Sigmoid arteries**

supplies the sigmoid colon (mesosigmoid)  
2 -4 branches

## **Superior rectal artery**

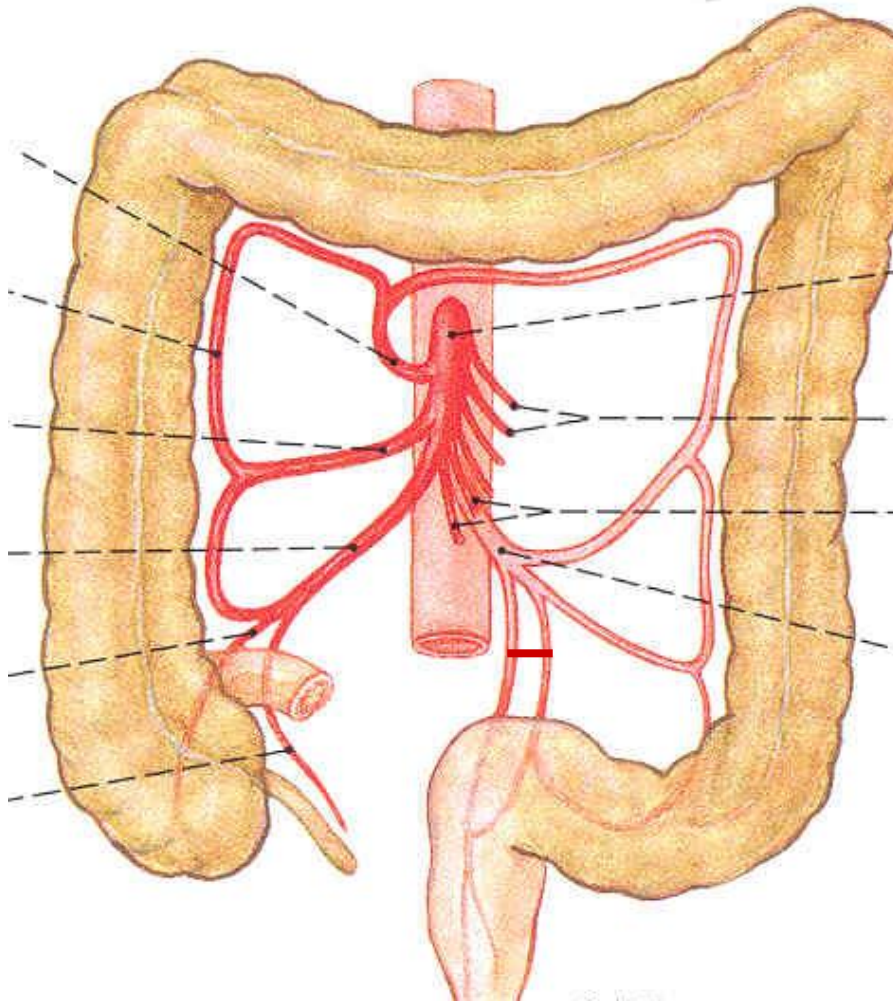
Supplies the upper (larger part of the rectum (anastomoses with the middle rectal arteries – branch of the internal iliac)



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# IMPORTANT ANASTOMOSES



## **ARCUS RIOLANI MAJOR (HALLER)**

***Drummond arch***

*secondary anastomosis*

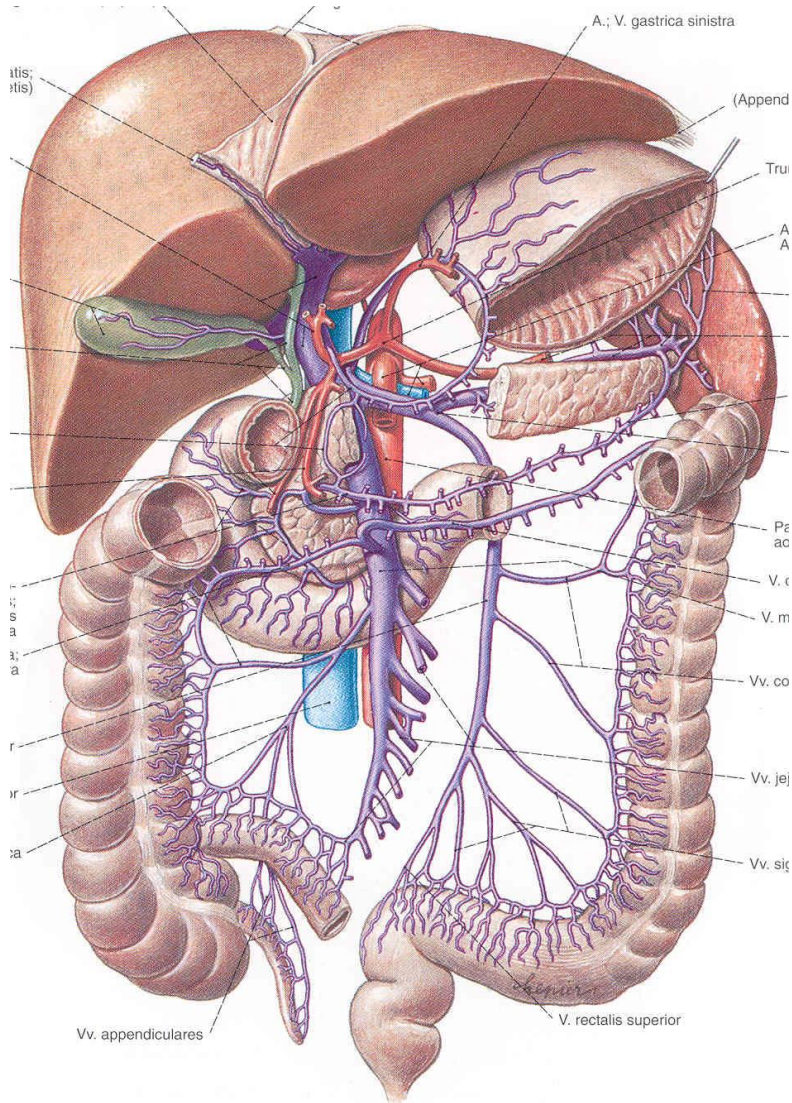
**Connects the middle and left  
colic arteries (*embryological  
border*)**

## **SUDECK'S POINT**

***connects the superior rectal  
artery and the last sigmoidal  
branch***

***Anastomosis between the  
superior rectal and middle  
rectal arteries***

# VENOUS DRAINAGE



Veins follow the course of arteries and aim for the portal vein

## **SUPERIOR MESENTERIC VEIN**

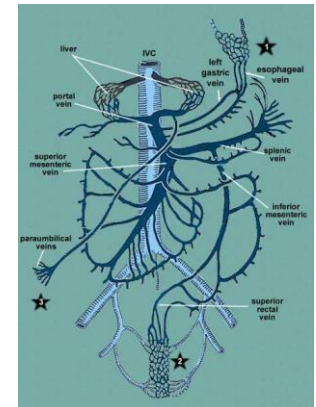
*Drains the ileocolic, right and middle colic veins*

*Opens directly in the portal vein*

## **INFERIOR MESENTERIC VEIN**

*Drains the left colic, sigmoid and superior rectal veins, opens either directly in the portal vein or indirectly, via the splenic vein*

***Please note the perirectal PORTOCAVAL ANASTOMOSIS***

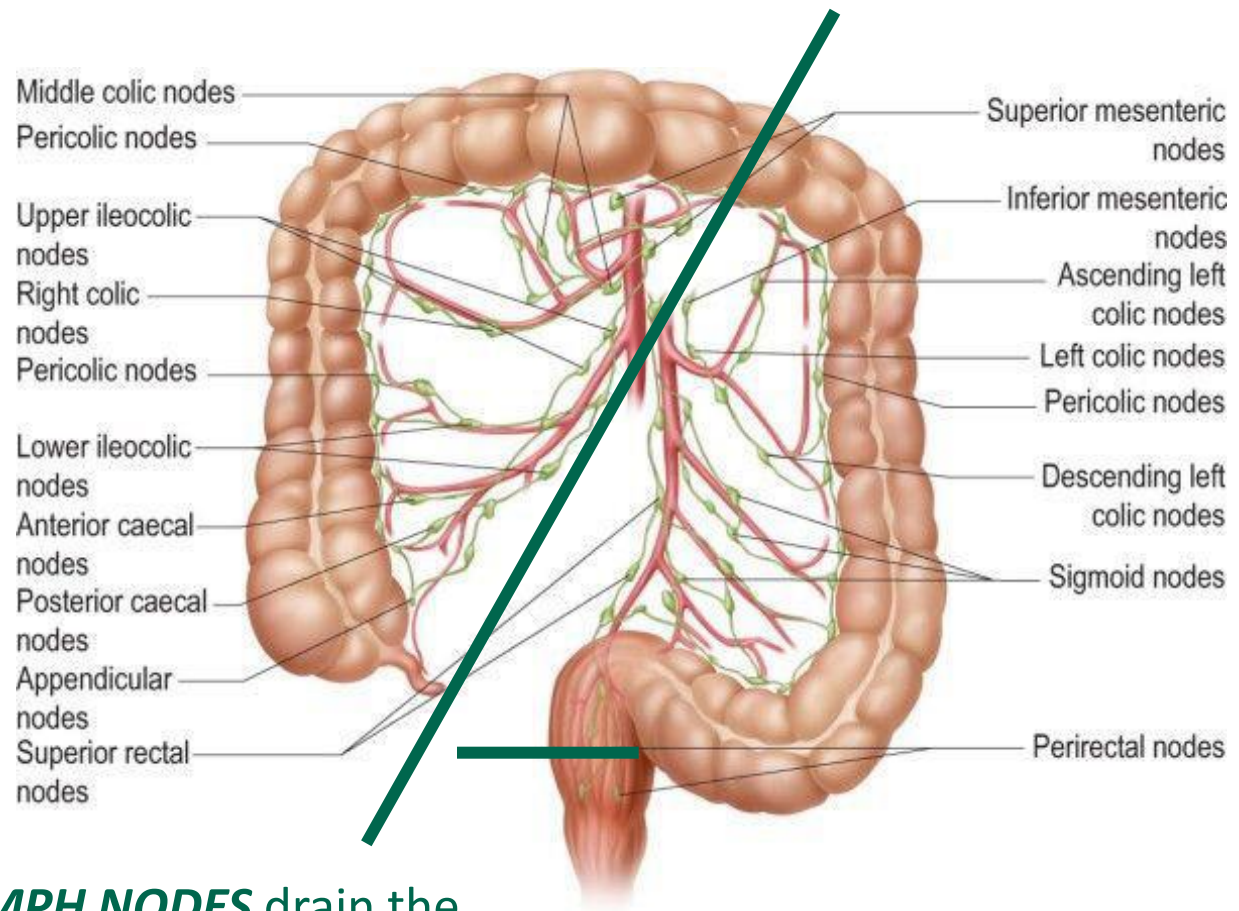




# LYMPHATIC DRAINAGE

Lymph vessels run with the arteries

**SUPERIOR MESENTERIC LYMPH NODES** drain the cecum, ascending colon, transverse colon via the right and middle colic lymph nodes



**INFERIOR MESENTERIC LYMPH NODES** drain the *descending colon via the left colic lymph nodes*; the sigmoid colon via the sigmoidal lymph nodes; the upper 1/3 of the rectum

**INTERNAL AND EXTERNAL ILIAC LYMPH NODES** drain the inferior 2/3 of the rectum

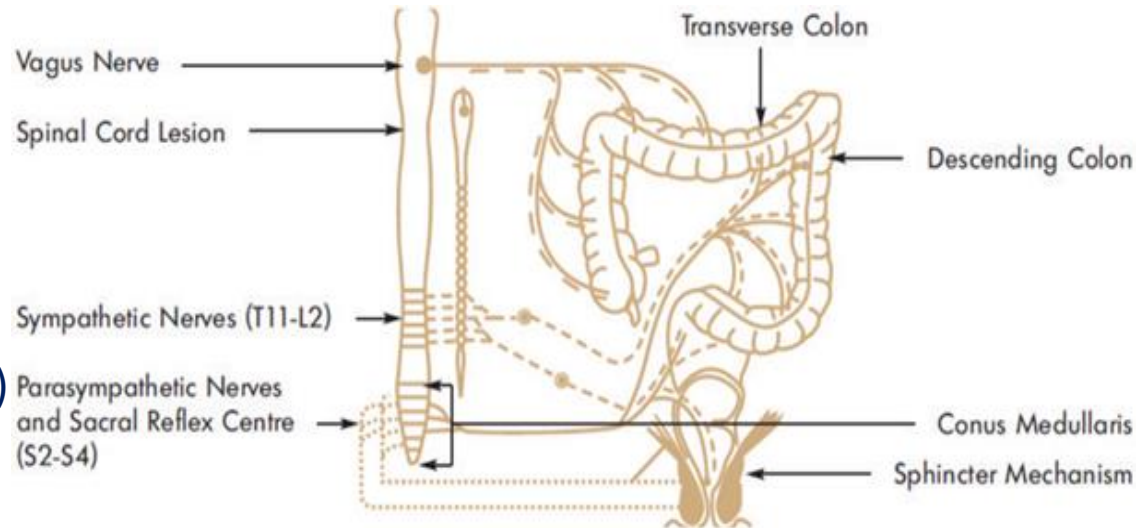


# INNERVATION

**PARASYMPATHETIC INNERVATION**  
(secretomotor and motor to the colic muscles)

**Vagus nerve** innervates the colon to the splenic flexure (Cannon-Böhm's point).

**Sacral parasympathetic fibres** (S3-S5) innervate the descending colon, sigmoid colon and rectum.



**SYMPATHETIC INNERVATION**

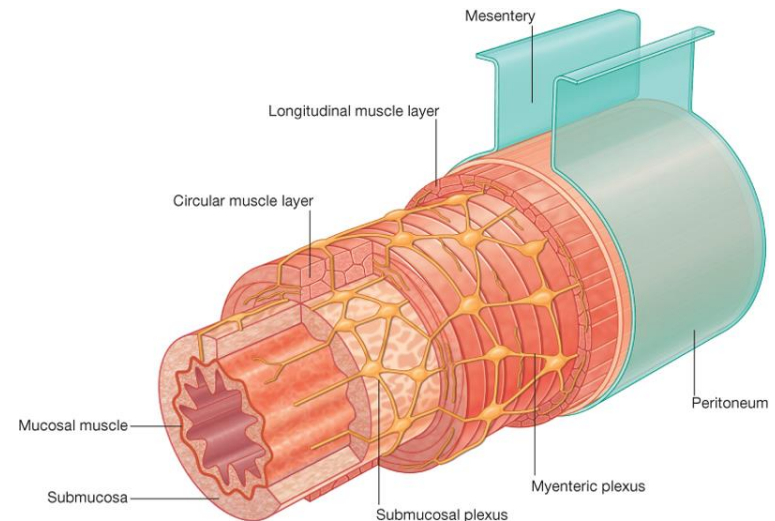
(vasoconstrictor and inhibitory to the colic muscles)

**superior and inferior mesenteric plexus**

**greater and lesser splanchnic nerves** (T6-11 ganglia)

**Enteric nervous system:**

**Meissner's plexus and Auerbach's plexus**



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